

University of the Armed Forces “ESPE”

Project's Name: UniStock

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Problem

The uniform and stationery store currently does not have a computerized mechanism that allows it to properly manage and control its inventory. Product information and stock quantities may be handled manually, making it difficult to know the exact number of uniforms and stationery materials available at any given time.

This lack of inventory control can cause problems such as inaccurate stock information, difficulty identifying products that are running low, errors when recording merchandise entries and sales, and unnecessary time spent checking products manually. It can also make decision-making more difficult because the store does not have reliable and centralized information about its merchandise.

Therefore, there is a need for a software solution that centralizes product information and automates the main inventory processes, allowing the store to maintain more accurate and organized stock control.

Overview

The Uniform and Stationery Store Management System is an object-oriented software project designed to support the administration of merchandise in a store that sells school or institutional uniforms and stationery materials.

The system will allow authorized users to register and classify products, consult product information, record purchases and inventory entries, register sales, make authorized stock adjustments, identify products with low stock, and generate inventory reports. It will also include user and role management to control access to the system's functions.

The main purpose of the system is to replace or reduce manual inventory control with a centralized and structured solution. By automatically updating stock when merchandise enters or leaves the store, the system will help reduce errors and provide more reliable information for daily operations.

Background

Inventory management is an important activity for businesses that sell physical products because it allows them to know what merchandise is available, what products need replenishment, and how products move through the business. When these processes are performed manually, maintaining accurate information can become increasingly difficult as the number of products and transactions grows.

A uniform and stationery store manages different types of merchandise, including clothing items such as shirts, pants, jackets and sportswear, as well as materials such as notebooks, pens, pencils, folders and other school supplies. These products can have different prices, categories and stock levels, which makes an organized inventory structure necessary.

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From a software engineering perspective, Object-Oriented Programming provides an appropriate approach for modeling this type of system because the real-world elements involved in the business can be represented as objects. Products, users, sales, inventory movements and reports can be represented by classes with attributes and behaviors. Concepts such as encapsulation, inheritance, abstraction and polymorphism can therefore be applied in the development of the project.

The proposed system will serve as an academic OOP project while also representing a practical solution to the store's inventory-control problem. Its design can be expanded in the future with additional features such as suppliers, customers, payment methods, barcode scanning, purchase orders and more advanced sales reports.